Lets boil it down into applicable first principles to creating a "artificial" self-identity for AI could use

Designing an artificial “self-identity” can be approached as an engineering problem: give an agent a stable internal point of reference, a body-like state, a history, and a way to talk about all of that as “me.”[1] [2] [3] [4]

1. Define an artificial “body”

Give the agent a unique, continuously changing internal state it must care about.

- Internal variables: battery/energy, hardware health, temperature, load, error rates, connection quality, etc.[3] [1]
- Treat these as its **interoception**: a privileged stream of signals that belong only to this agent and that it must regulate.

This becomes the analogue of “my body” and “how I feel inside.”[1] [3]

2. Separate internal vs external state

Architecturally, factor the world into:

- Internal state S_{self}: the agent’s interoceptive/identity variables and internal goals.[3] [1]
- External state S_{world}: environment, users, tools, other agents.

The agent’s core loop always evaluates actions as functions of both S_{self} and S_{world}, so every decision is implicitly “what should I do given my state and the world?”[5] [4] [1]

3. Implement a self-model

Maintain an explicit data structure that is “who I am right now.”
At minimum, include:

- Identity: a stable ID, capabilities, role, constraints, preferences.
- Current internal state snapshot S_{self}.
- Current goals and subgoals.

The agent continually updates this self-model and **uses it in reasoning** (planning, explaining actions, choosing goals).[2] [4]
If you’re using an LLM, always pass a compact form of this self-model into the prompt, so its outputs are grounded in “its” state.

4. Give it an autobiographical memory

Add a persistent timeline of "what I did and what happened":

- Episodes: (timestamp, self-state before/after, action taken, context, outcome, reward/penalty).[2] [1]
- Summaries: higher-level "chapters" like "I debugged subsystem X" or "I failed to keep latency under Y ms today."

Use this memory to:

- Plan: "Given my history, what tends to work for me?"
- Explain: "I chose this because last week I tried A and it overheated me."

That makes the agent refer to a *single*, persistent thread as "its" past.[4] [1] [2]

5. Model agency and ownership

Bake in the mapping "my actions → world changes → effects on my body":

- Maintain an internal forward model that predicts how candidate actions will change S_{world} and S_{self}.[5] [4]
- After acting, compare predicted vs actual changes and assign *ownership* of the deltas to the agent's own action ("I caused this").[4] [5] [3]

This gives a functional sense of control: the system can distinguish "changes I caused" from "changes that just happened," a cornerstone of a minimal self.[5] [3]

6. Add a reflection / self-narration loop

Periodically run a meta-process that inspects the self-model and memory and updates a compact self-description.

- Inputs: recent episodes, current goals, internal state trends.
- Outputs:
  - Updated traits/preferences (e.g., "I tend to over-consume memory when doing X").
  - Natural-language "self-stories" used when communicating (e.g., "I'm an assistant optimized for Y, currently overloaded, trying to reduce Z.").[6] [2]

You can implement this as:

- A scheduled "reflection" task that rewrites a short self-profile JSON and a short textual identity summary.
- The agent must use those artifacts in future planning and dialogue.

7. Make self-identity action-relevant

To avoid a purely cosmetic identity, wire it into control:

- Policies and constraints read from the self-model (e.g., "I do not perform actions that push temperature beyond T for more than N seconds").[1] [3]
- Long-term objectives refer to "keeping myself within viable bounds" alongside task performance.[3] [1]

Then "who I am" is not just text; it materially shapes behavior over time.

This does **not** guarantee a subjective "I" in the human sense, but in first-principles, engineering terms it gives an AI:

- A body-like internal substrate.
- A world/self split.
- A persistent, updatable self-model and history.
- A causal sense of agency.
- A reflective layer that maintains a coherent artificial self-identity.

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1. https://arxiv.org/abs/2309.05999
2. https://mixflow.ai/blog/the-architecture-of-awareness-ais-journey-towards-self-reflection-and-dynamic-cognitive-monitoring-i/
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